

VoxShield Edge

Real-Time, Privacy-Preserving Voice-Clone Defense for Indian Telephony

“When a cloned voice can sound like someone you trust, the device needs to know the difference.”

5.9% EER · deployed fusion (held-out In-the-Wild)	6.8% EER · unseen-generator deepfakes (OOD, MLAAD)	0.98 ROC-AUC on OOD deepfakes	45–80 TOPS Hexagon NPU (Snapdragon X / X2)
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Evidence labels used throughout: [VERIFIED] · [PROJECT RESULT] · [IMPLEMENTED] · [EXPERIMENTAL] · [PROPOSED] · [TARGET] · [ILLUSTRATIVE]

Individual submission · Devansh Goenka · GitHub: DG10911/voxshield · Working system at TRL-5 [PROJECT RESULT]

1 · Executive Summary

VoxShield Edge is an on-device AI security layer that detects AI-generated and cloned voices during telephony conversations and explains *why* — running locally on a Snapdragon-powered HP PC so sensitive call audio need not leave the device. It is **telephony-first** (8 kHz G.711), **Indic-aware** (10 Indian languages), **multi-detector** (five complementary models + learned fusion), **streaming** (a verdict within the first seconds), **explainable**, and **non-destructive** (recommends step-up verification; never auto-blocks). [IMPLEMENTED]

Measured evidence [PROJECT RESULT]: 5.9% EER on a held-out in-the-wild corpus; 6.8% EER / 0.98 AUC on out-of-distribution deepfakes from unseen generators; Indic fake recall 42% → 82% after Indic-aware retraining; genuine-Indic false positives 11.7% → 6.3%. **Snapdragon status**: the detector runs today on CPU/GPU; the NPU port via ONNX Runtime QNN + Qualcomm AI Hub is a defined optimisation roadmap [PROPOSED] — no on-NPU latency is claimed until measured.

2 · Problem & Indian Telephony Threat Model

Voice cloning is now cheap and fast. In India, phone-based fraud — fake bank officials, “digital arrest” scams, relative-in-distress calls — increasingly uses synthetic voices in Indian languages over ordinary 8 kHz phone lines. The attack chain:

Attacker → cloned voice of a trusted person → victim over a phone call → social-engineered financial fraud.

- Cloning needs only seconds of reference audio; scripts are localised per language.
- Telephony (G.711, 8 kHz) strips the high-frequency detail many detectors rely on.
- Sensitive customer audio should not be shipped to a foreign cloud (data-governance).
- A verdict *after* the call ends is forensics, not protection — detection must be real-time.

3 · Why Voice-Clone Detection Is Hard Here

Dimension	Common assumption	This setting
Channel	Studio / wideband	8 kHz G.711 narrowband, codec loss, noise
Language	English-centric data	Indic phonetics; code-switching
Attack coverage	Single detector / known generators	Diverse, evolving TTS & voice-conversion
Output	Black-box score	Operational risk + reason for an analyst
Deployment	Cloud inference	On-device / controlled infrastructure

Why the specifics matter: **8 kHz** is the real telephony rate; **G.711 μ -law** is the dominant codec, and both discard cues above ~3.4 kHz. A **3-second window** gives enough context to judge a voice; a **1-second hop** yields an early verdict that refines during the call. **Multiple detectors** reduce single-model fragility; **Indic-aware** handling prevents both missed Indic fakes and over-flagging genuine Indian speakers; **low false positives** matter because over-flagging real customers is costly.

4 · VoxShield Edge — Solution & Product Experience

An endpoint application on an HP Snapdragon AI PC. Audio is processed locally; VAD gates speech; streaming windows feed five specialist detectors; an adaptive fusion layer combines the evidence; the user sees a risk verdict and the reasons behind it. No automatic blocking occurs. [\[IMPLEMENTED\]](#)

VOXSHIELD EDGE — LIVE CALL ANALYSIS
Language: Hindi Signal: 8 kHz telephony Window 3.0s / Hop 1.0s
Acoustic [OK] SSL [!] Indic [!] Prosody [OK] Cepstral [!]
FUSION RISK 91% -> LIKELY SYNTHETIC VOICE
Evidence: synthetic spectral · abnormal prosody · vocoder-like energy · detector agreement
Processing: LOCAL DEVICE Internet required: NO

(UI is an illustrative concept; a working streaming console is included in the prototype materials.)

5 · System Architecture

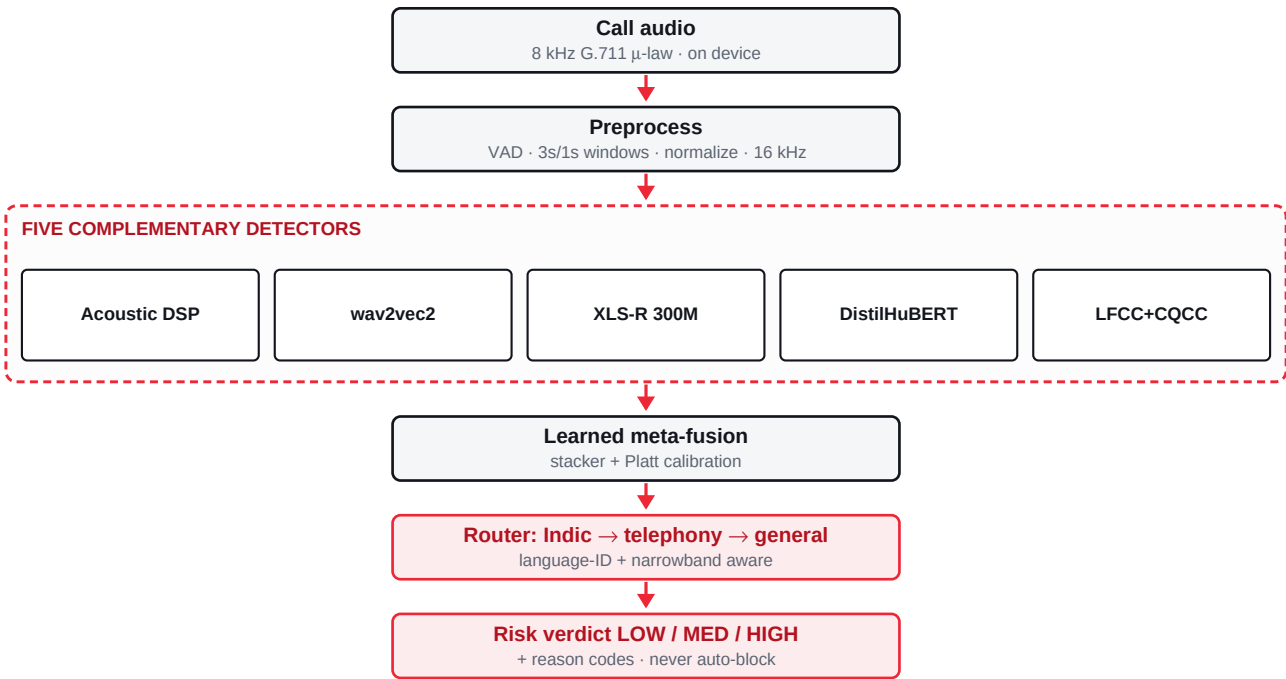


Figure 1 — On-device architecture: audio → preprocessing → five detectors → learned fusion → routing → explainable risk. [\[IMPLEMENTED\]](#)

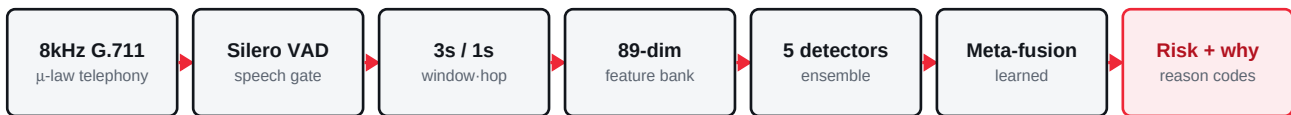
6 · Detector Architecture & Fusion

VoxShield mixes detector families on the hypothesis that they respond differently to synthesis artifacts and channel distortion, so fusion improves robustness. The 89-dimensional feature bank (LFCC, CQCC, group-delay phase, F0, jitter, shimmer, and a >6 kHz band computed at a 16 kHz analysis rate) feeds classical detectors; self-supervised models (wav2vec2, XLS-R 300M, DistilHuBERT) add learned representations; a learned meta-stacker combines them. [\[IMPLEMENTED\]](#)

Configuration	EER	Reading
Best single detector (XLS-R 300M)	16.0%	strong but fragile
Naive averaging	19.1%	worse than best single
Learned meta-fusion	5.9%	the fusion produces the gain

The learned stacker — not any single model — produces the performance gain [\[PROJECT RESULT\]](#). Naive averaging is worse than the best single model, which is why a *learned* fusion matters.

7 · Streaming Pipeline



Streaming: a verdict forms within the first seconds and refines as the call proceeds

Figure 2 — Streaming inference pipeline (3s window · 1s hop). [IMPLEMENTED]

8 · Indic-Language Intelligence

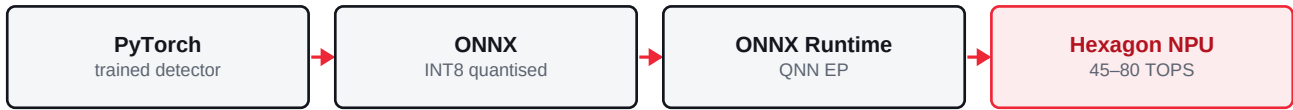
A multilingual backbone does not by itself make a detector language-robust: synthesis artifacts and genuine-speaker distributions shift across languages. VoxShield adds an Indic-aware detection path with on-device language-ID routing (Indic → telephony → general) and retrains on Indic material. Measured on the tested Indic conditions [PROJECT RESULT]:

Metric	Before	After	Scope
Indic fake recall (MMS-TTS)	42%	82%	single-generator, tested set; English EER unchanged
Genuine-Indic false-positive rate	11.7%	6.3%	channel-aware; fewer genuine-customer flags

Supported languages (target): Hindi, Tamil, Telugu, Bengali, Marathi, Kannada, Malayalam, Gujarati, Punjabi, English. [TARGET]

9 · Snapdragon Optimisation

Snapdragon is core to the product, not a label. VoxShield Edge targets the Hexagon NPU for power-efficient, offline, sustained inference on an HP Snapdragon AI PC.



Compiled & profiled on real Snapdragon devices via Qualcomm AI Hub (qai-hub)

Figure 3 — On-device inference path (proposed optimisation). [PROPOSED]

Layer	Approach	Status
Hardware	Snapdragon X Elite/Plus (45 TOPS NPU); X2 Elite (80 TOPS)	[VERIFIED]
Inference runtime	ONNX Runtime + QNN Execution Provider (onnxruntime-qnn), Windows/ARM64	[PROPOSED]
Compile / profile	Qualcomm AI Hub (qai-hub) on real Snapdragon devices	[PROPOSED]
Optimisation	INT8 quantisation; heterogeneous CPU/GPU/NPU placement	[PROPOSED]
Ecosystem models	WavLM-Base-Plus backbone + Whisper (both on AI Hub)	[VERIFIED]

Honest note: not every model is claimed to run on the NPU today. wav2vec2/XLS-R are not pre-packaged on AI Hub and, like Qualcomm's own edge Whisper port (which adapted attention/linear ops), may need adaptation for full NPU offload — validated via AI Hub profiling. This is the optimisation roadmap, not a measured result.

10 · Privacy / On-Device Architecture

CLOUD DETECTION (typical)



audio leaves the device · bandwidth · latency · privacy exposure

VOXSHIELD EDGE (this project)

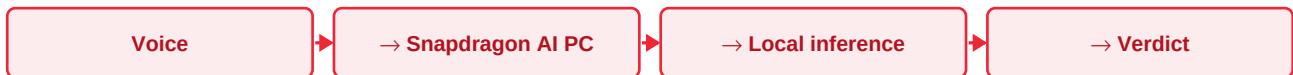


Figure 4 — Cloud path vs on-device path. [ILLUSTRATIVE]

- Raw call audio is processed locally — architected so it need not leave the device. [IMPLEMENTED]
- Offline operation: works air-gapped / in low-connectivity branches. [TARGET]
- Latency: streaming windows form an early verdict; on-NPU latency to be measured. [PROPOSED]
- Power efficiency: sustained NPU inference targeted for all-day endpoint use. [TARGET]

11 · Evaluation Methodology & Results

Detectors are trained on some corpora and evaluated on held-out / out-of-distribution data the models did not see in training — a deliberately hard generalisation test. EER (Equal Error Rate) is the operating point where false-accept and false-reject rates are equal (lower is better); it is not accuracy. In-corpus and out-of-distribution figures are reported separately and never conflated.

Evaluation	Metric	Result	Label
Held-out in-the-wild (fusion)	EER	5.9%	[PROJECT RESULT]
Unseen-generator deepfakes (MLAAD, OOD)	EER / AUC	6.8% / 0.98	[PROJECT RESULT]
Indic fake recall (MMS-TTS)	recall	42% → 82%	[PROJECT RESULT]
Genuine-Indic false positives	FPR	11.7% → 6.3%	[PROJECT RESULT]
On-NPU latency / power	ms · W	to be measured	[PROPOSED]

12 · Explainability

Each verdict carries reason codes derived from detector outputs and the feature bank, so an analyst sees *why* a call scored high — spectral inconsistency, vocoder-like >6 kHz energy (wideband only), phase / group-delay irregularity, missing prosody (F0/jitter/shimmer), absence of natural breath, and detector disagreement. These are signal-derived indicators, not formally faithfulness-tested explanations. [\[IMPLEMENTED\]](#)

13 · Competitive Positioning

Capability	Cloud voice detection	Single-model deepfake	Generic audio deepfake	VoxShield Edge
Telephony (8 kHz) optimised	Varies	Varies	Rarely	Designed for
Indic-language aware	Varies	Rarely	Rarely	Designed for
Streaming / early detection	Varies	Varies	Post-hoc	Designed for
Multi-detector fusion	Varies	No	Varies	Demonstrated
Explainable evidence	Varies	Rarely	Varies	Demonstrated
On-device / offline	No	Varies	Varies	Targeted
Privacy (no audio egress)	No	Varies	Varies	Designed for

Language is deliberately conservative — “Designed for / Demonstrated / Targeted” — and “Varies” where a general claim cannot be evidenced across products.

14 · Evidence Matrix

Claim	Evidence	Status	Demonstration
Real-time detection	3s window / 1s hop streaming	Implemented	Live streaming audio
Multi-detector fusion beats single	16.0 / 19.1 / 5.9 EER scoreboard	Project result	Result table
Generalises to unseen generators	6.8% EER / 0.98 AUC on MLAAD (OOD)	Project result	Held-out eval
Indic-aware helps	recall 42→82; FP 11.7→6.3	Project result	Before/after eval
On-device / private	local pipeline, no audio egress	Implemented (CPU/GPU)	Offline demo
Runs on Snapdragon NPU	ONNX→QNN→Hexagon path + AI Hub	Proposed	AI Hub profiling (next)
Explainable verdict	reason codes from detectors	Implemented	Evidence panel

15 · Limitations

- Unseen generators / codecs can evade until incorporated; cross-generator remains a validation target.
- Speaker-disjoint and real captured-PSTN evaluation are not yet complete.
- Very short segments, heavy noise and code-switching reduce reliability.
- On-NPU latency/power are not yet measured; NPU offload of transformer detectors may need adaptation.
- In-corpus figures reflect corpus separability; the honest generalisation number is the OOD 6.8% EER.

16 · Roadmap & Adoption

Horizon	Milestone	Label
Now	ONNX export + INT8 quantisation; AI Hub compile/profile on Snapdragon X/X2	[PROPOSED]
Next	Measured on-NPU latency + power; Windows endpoint app; streaming demo	[TARGET]
Then	More Indic languages; larger real-world corpus; challenge-response liveness	[TARGET]
Product	Contact-centre integration; enterprise SDK/API; telecom pilots	[PROPOSED]

Adoption: banks, NBFCs and contact centres that need Indic-language, telephony-quality, private fraud screening on endpoints — a Snapdragon AI PC per fraud-desk analyst, no per-call cloud cost.

17 · Conclusion

VoxShield Edge is a technically serious, evidence-driven edge-AI security prototype for a real Indian problem, with a credible path to Snapdragon deployment. It detects early, explains clearly, and is designed to protect privately — on the device.

Detect early. Explain clearly. Protect privately.

GitHub: DG10911/voxshield · Pitch deck (PDF + PPTX), prototype console, and one-page summary submitted alongside · Contact: devanshgoenka03@gmail.com